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Stand-alone microprocessor controlled fast sweep Langmuir probe driver

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This article describes a power supply and data logger for a Langmuir probe interfaced to a personal computer. The system provides a voltage sweep range from −125 to +100 V in 100 steps in a time ~0.15 s. Restricted sweep ranges and single point operation are also possible. Probe current measurements are in the range from −1.0 to +100 mA with a precision of 5 μA on the most sensitive range, while the voltage may be set with a precision of 56 mV. Novel features of the system include: the use of integrate-and-dump techniques to implement the process of analog-to-digital conversion and to provide effective noise suppression; a solution to the problem of floating the power supply on top of the potential developed across the grounded current sensing resistor based on the power supply rejection ratio characteristics of a high voltage operational amplifier; and the development of an interface and control board employing the GPIB protocol to communicate with a host computer. Successful operation of the system has been demonstrated in the electrically noisy environment of a helicon plasma source. © *1997 American Institute of Physics.*
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I. INTRODUCTION

In processes that use plasmas as the reactive medium (for example in integrated circuit manufacture), the plasma parameters, electron temperature, and electron density, often need to be monitored closely as these two parameters, among others, are usually critical in determining the quality and speed of the process. The Langmuir probe is used extensively in such devices.

The Langmuir probe is a seemingly simple device that in its most basic form is merely a small piece of metal inserted into a plasma. A potential is imposed upon the metal probe relative to a ground that is in contact with the plasma. The current driven by this potential is measured for a range of voltages, both negative and positive. From the shape of the current–voltage ($I-V$) characteristic, the electron temperature and density can be deduced. Not only is this a very simple technique, it can be used over a wide range of plasma conditions.[1] Unfortunately this simple device is fraught with problems in the interpretation of the $I-V$ characteristic. Different plasma conditions impose different interpretations. For instance, the presence of either a magnetic field or a radio frequency field considerably increases the complexity of the theory.[2]

This article is concerned with the mechanism of collecting the data for the $I-V$ characteristic as rapidly and accurately as possible under computer control remembering that these probes are often used in hostile environments with high power radio frequency sources close by (used to generate and heat the plasma). The interpretation of the data collected depends to a large extent on the plasma's environment and is not discussed here.

The $I-V$ characteristic of the probe is obtained by sweeping the probe potential from about −100 to +100 V on a time scale short compared with the time scale of any changes within the plasma. Since the $I-V$ data must undergo extensive analysis and interpretation to extract the desired parameters, computer control and data acquisition are essential.

In this article, we describe the development of a fully automated Langmuir probe driver based on a previous design published by Duffey, Cheetham, and Rayner.[3] The new driver sweeps the probe potential from −125 to +100 V in up to 100 steps with a sweep period of less than 200 ms. If detail is required, the range can be restricted and up to 100 points taken in the defined range. The probe current and potential are measured at each point and the data stored in on-board random access memory (RAM) to be down loaded to the central computer on request. The limit of 100 points is due only to the amount of RAM currently available. The technique of matched filters (integrate-and-dump) has been implemented in the measurement circuit to maximize the signal-to-noise ratio (SNR) and provide internal analog-to-digital conversion (ADC) while averaging the noisy data over the step interval; a technique used extensively in communications systems. This technique avoids the necessity of taking many readings using a fast ADC and averaging in order to reduce noise.[3] Gain switching within the unit is automatic and optimized for maximum noise immunity leading to a resolution of 5 μA in 1 mA.

In brief, the specifications for the new system are as follows:

(1) Linear voltage sweep:	Adjustable from $V_A=-125$ to $V_F=+100$ V or subranges between these limits.
(2) Modes:	Sweep in up to 100 steps, or single point acquisition.
(3) Sweep time:	$\Delta t \leqslant 0.15$ s.
(4) Current detection:	From $I_i=-100$ to $I_e=+100$ mA $\Delta I \leqslant 0.5\%$.

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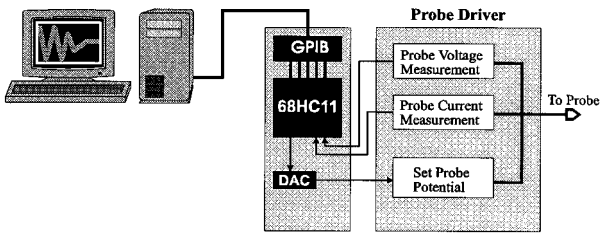


FIG. 1. Block diagram of the probe system.

(5) Robust operation: Ability to operate in noisy electrical environments. In particular, immunity to 13.56 MHz intrusion.

(6) Interface GPIB to host computer.

II. THE PREVIOUS DESIGN

The previous Langmuir probe driver[3] was controlled and logged using a commercial data acquisition card installed in a PC. It was designed and constructed to sweep a Langmuir probe from -125 to $+100$ V in 4000 steps with a voltage setting and sensing resolution of 56 mV, and current determination in the range of -1.0 to $+100$ mA with a resolution of 10 μA on the most sensitive range. The system has been in use for some time and has worked reliably within its specifications. The sweep time for 4000 points is 160 ms allowing 50 sweeps to be recorded in less than 8 s. Thus, averaged data is produced at a rate of 2 ms per point.[3]

The design of this driver had four main disadvantages; all of which have been addressed in the new design presented in this article:

(1) The circuit suffered from an undue amount of noise on the current measurement requiring many sweeps ($\sim$50) to be averaged for a reasonably smooth $I-V$ trace to be obtained.
(2) The power supply floated on top of the potential developed across the current sensing resistor. This meant that a complex feedback arrangement was necessary so that the ground referenced DAC could drive the power supply.
(3) The structure of the circuit led to a substantial input bias current that had to be offset in software.
(4) Since all control was derived from the controlling PC, that machine was necessarily completely occupied while the probe was in operation.

III. PROBE DRIVER SYSTEM

Figure 1 shows the probe driver system in block form. A GPIB link connects the controlling PC to an interface board based on the 68HC11 microcontroller. The interface board controls the power supply which consists of three main modules as shown in Fig. 2. The **power supply block** accepts a control voltage (provided by the DAC on the interface board) which sets the voltage applied to the probe. The **current measurement block** detects the voltage developed by the probe current passing through a current sensing resistor using an integrate-and-dump (I&D) network. The **voltage measurement block** measures the probe potential, again using an integrator, and returns the value to the PC via the interface board.

A. The integrate-and-dump network

For the I&D network, at the beginning of the measurement cycle, the dump switch, which closed at the end of the previous cycle to discharge the capacitor, opens. At the same time, a counter/timer on the interface card is initiated. For an input voltage V_i, the output V_o increases linearly according to the relation:

$$V_o = V_i t/\tau, \tag{1}$$

where $\tau = RC$ is the time constant of the integrator shown in Fig. 3. At some time T, the output reaches a level V_{ref} which causes the comparator to change state. This signal passes through the digital opto-isolator to the interface board and stops the counter/timer. The stored count is proportional to the integration period T which in turn is related to the input voltage by the expression

$$V_i = V_{\text{ref}}\tau/T.$$

There are several major advantages of the integrate-and-dump circuit compared with a conventional ADC-based approach in that it:

(1) enhances the signal-to-noise ratio (SNR),
(2) carries out the process of analog-to-digital conversion,
(3) permits simple data analysis to calculate the values of V and I, and
(4) allows simple opto-isolation as digital, rather than analog, signals are involved.

To achieve similar performance a high speed ADC taking multiple samples per step would be required.

While a dual-slope integration technique may be desirable as it produces a count that is directly proportional to the voltage, this was not implemented as it would significantly increase the data acquisition time.

B. Range changing

As probe currents of interest have a dynamic range of 80 dB from around 10 μA to 100 mA, some form of range changing is required to keep the integration period T within reasonable bounds. This has been achieved by using a set of current sensing resistors arranged in 20 dB steps with the software selecting the most suitable resistor to give a reasonable value of T for a given current. If the sense resistor is too small for a given current then an unduly long integration period is required. Under these conditions, the valid result is stored and the range adjusted to give a shorter integration time by increasing the sense resistor ready for the next sample. The system also has current and voltage over-range flags which are checked to see whether either signal is too large. If so, then the range(s) change automatically until the flag(s) clear and the now valid results can be recorded.

The clock period of the counter/timer is 0.5 μs and the minimum integration period is 100 μs thus leading to a worst

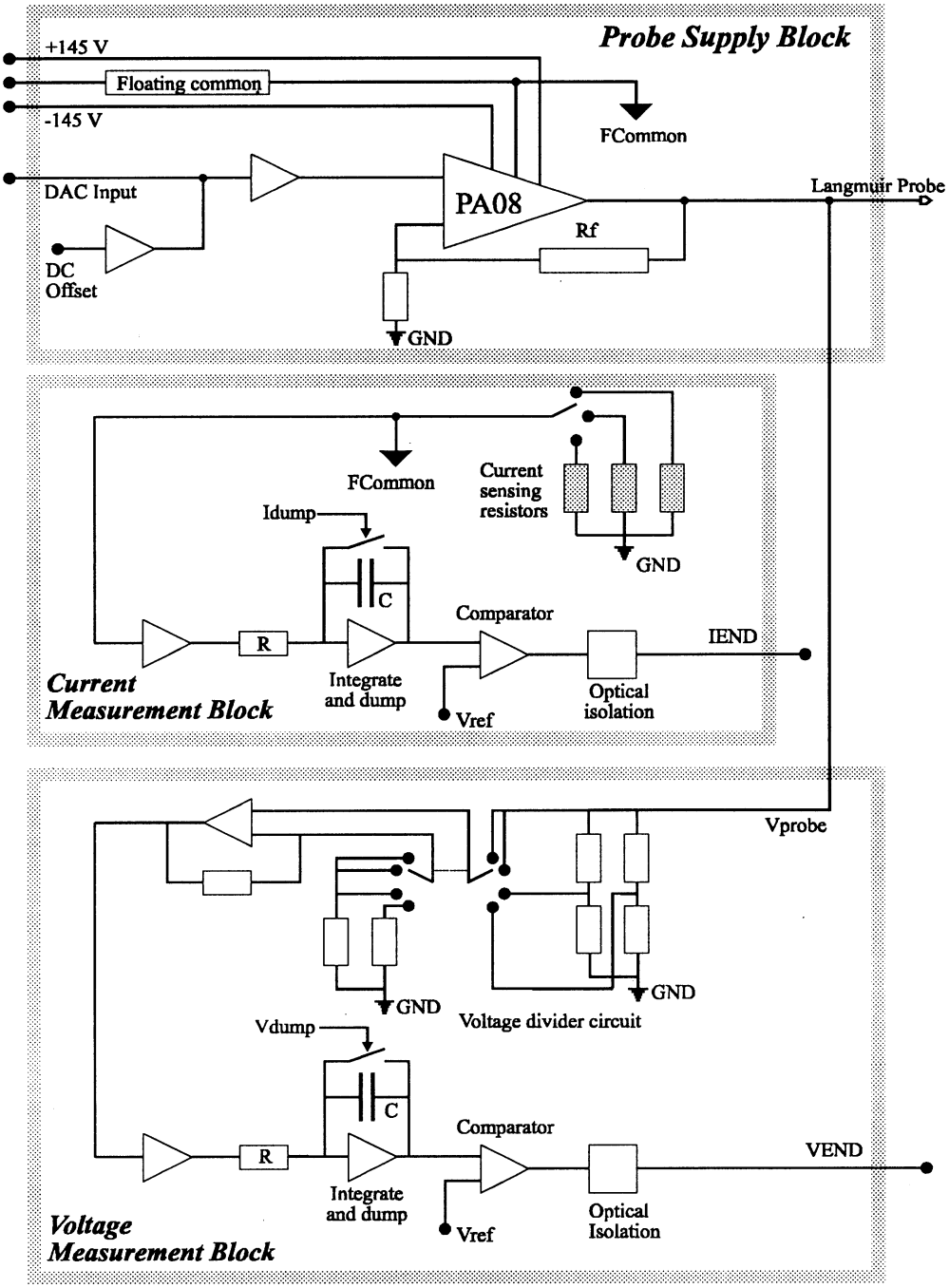


FIG. 2. Detailed diagram of the Langmuir probe driver.

case precision of 0.5% in the current measurement. For the maximum period of 1 ms, the precision improves by a factor of 10.

The sweep rate tends to slow down in the vicinity of zero current as the system spends some time searching for the correct range.

The **voltage measurement block** covers a 68 dB dynamic range of voltages and again employs the integrate-and-dump technique and range switching of the voltage divider sampling the probe voltage.

The voltage divider consists of a pair of high value resistors to minimize circuit loading. However because of the stray capacitance shunting the larger resistor, and the overshoot which results from step changes in the probe voltage, a compensation capacitor is required in parallel with the smaller resistor. In addition, a small software delay is included so that the integration process only starts after the voltage level has stabilized.

For the voltage measurements, an appropriate range is set based on the *a priori* knowledge of what voltage should exist as set by the DAC.

C. The floating power supply

In previous designs of Langmuir probe drivers, numerous attempts have been made to decide whether it is better to float the current sensing resistor on top of a grounded power

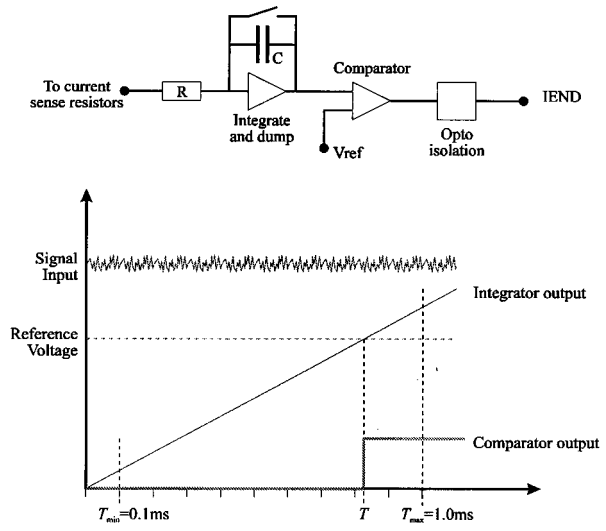


FIG. 3. (a) Integrate-and-dump circuit. (b) Timing for the integrate-and-dump.

supply[4] or to ground one side of the sense resistor and float the supply.[5,6] Many arguments for one or other schemes have been advanced and ingenious solutions developed.[7]

In the present case, it was decided to ground the sense resistor as it vastly simplified the measurement of the probe current but complicated the design of the power supply. As shown in Fig. 2, the heart of the power supply is a high voltage, high current operational amplifier, the PA08 manufactured by Apex Microtechnology Corporation. The PA08 is configured as a noninverting amplifier with the feedback network, and hence the input signal, **referenced to ground**. The noninverting arrangement significantly reduces the input current which can be a major problem if an inverting amplifier is used.[3] Referencing the input to ground greatly simplifies the interface of the amplifier to the DAC.

The PA08 is powered by a **floating** +/− 140 V power supply. The floating common between the positive and negative supplies is connected to the top of the current sensing resistor. The current from the probe therefore passes through the PA08 to one or other of its supply rails and through the current sense resistor to ground as shown in Fig. 4. With up to 10 V appearing across the sense resistor, the power supply voltages to the PA08 vary between 130 and 140 V. However the power supply rejection ratio (PSRR) of the PA08 is >80 dB over the frequency range of interest giving a 1 mV change in the input signal which, with a gain of 10 in the present application, leads to an error in the set voltage to the probe of 10 mV. This worst case error is small compared with the minimum resolution of 56 mV in our system. The small component of the current from the probe that passes through the amplifier's feedback network directly to ground, and is therefore not detected by the sense resistor, is compensated for in the software.

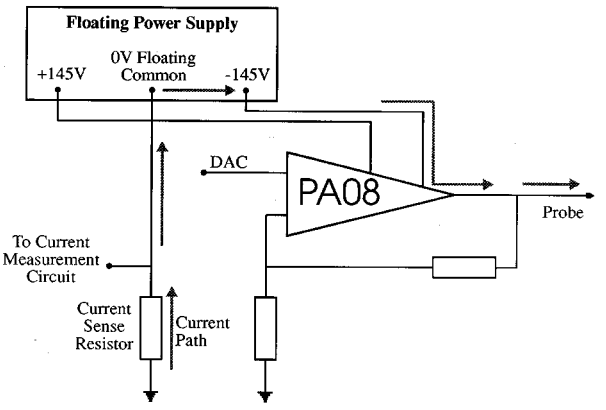


FIG. 4. Circuit for the PA08 high voltage operational amplifier showing the probe current path through the floating power supply to the current sense resistor.

D. Shielding and grounding

As the system must work in an electrically noisy environment, noise pick-up was minimized by placing the power supply in a fully enclosed steel cabinet. Line filters were installed in the mains power input and a toroidal mains transformer was employed. Gridded ground planes were employed on the circuit boards together with single point grounding in a star configuration. Internal partitions separated one part of the system from another.

IV. NOISE CONSIDERATIONS

Langmuir probes are often required to work in electrically noisy environments. Consequently an efficient system is required to enhance the signal-to-noise ratio of the measurements of the probe's current and voltage. Apart from low-pass filtering, one common technique is to take multiple current–voltage scans and to average the results. Alternatively multiple values of the current and voltage are recorded and averaged at each voltage level during a scan.[3,5,6]

The signal-to-noise ratio for such a system may be found as follows. If η is the single-sided noise power spectral density and the measurement system is represented by a single pole low pass filter of corner frequency f_c, the total noise power for a single sample is given by[8]

$$N_1 = \pi f_c \eta/2. \tag{2}$$

For n samples the total noise power N becomes nN_1. If V is the amplitude of the signal then for n samples the signal power is $(Vn)^2$. Thus the signal-to-noise ratio for n samples is given by

$$S/N = 2V^2 n/\pi\eta f_c. \tag{3}$$

In the present case, the integrate-and-dump provides excellent noise suppression through the continuous averaging provided by the integrator. As the integrator is essentially equivalent to a matched filter, its performance is close to optimum in terms of enhancing the signal-to-noise ratio.

For an integrator of time constant τ and integration period T, the signal power scales as $(VT/\tau)^2$ while the noise power goes as $\eta T/2\tau^2$. The resulting signal-to-noise ratio is given by[9]

$$S/N = 2V^2 T/\eta. \tag{4}$$

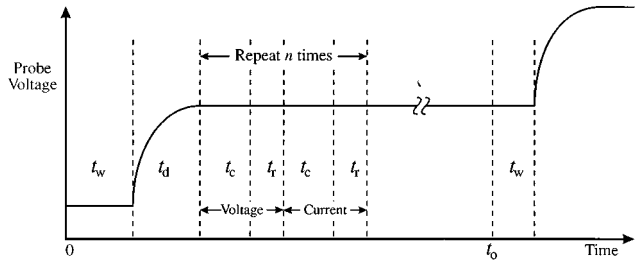


FIG. 5. Timing diagram for n samples at a given voltage level.

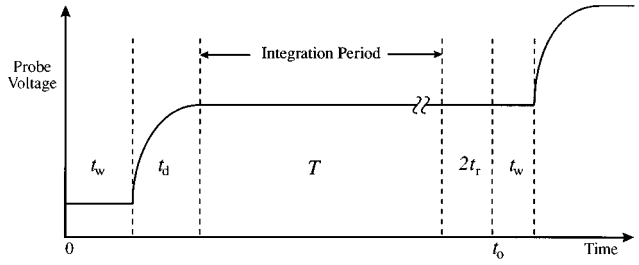


FIG. 6. Timing diagram for an integrate-and-dump network at a given voltage level.

Comparison of Eqs. (3) and (4) shows that the number of samples n required to match the performance of an integrator of period T is given by

$$n = \pi T f_c . \tag{5}$$

The choice of whether to use an integrator or to average over n samples depends on the characteristics of the measurement system.

Suppose that a system is required which collects samples at N_o different voltage levels in a total sweep time T_o where T_o is set by the time scale of changes in the plasma. The time available at each voltage level $t_o = T_o / N_o$. If we assume that the voltage level is set by a DAC with a write time t_w and that the system has a response time t_d ($= 1/f_c$ for a level within 0.2% of its final value) then, as shown in Fig. 5 for n samples at a given level:

$$t_o = t_w + t_d + 2n(t_c + t_r),$$

where t_c is the conversion time of the ADC reading the probe current or voltage, t_r is the time required to read the result into memory and the factor of two allows for reading both the current and voltage. This assumes that a single multiplexed ADC is employed. The number of samples that therefore can be acquired at each level is given by

$$n = \frac{(t_o - t_w - t_d)}{2(t_c + t_r)}. \tag{6}$$

For the integrator, Fig. 6 shows that

$$t_o = t_w + t_d + T + 2t_r ,$$

where it has been assumed that separate integrators have been employed for the current and voltage measurements and that only two reads are required to store the results of the integrations.

The time available for the integration T is therefore given by

$$T = t_o - t_w - t_d - 2t_r . \tag{7}$$

Thus given a system constrained by t_o and the characteristics of the microcomputer system employed to log the data, Eqs. (6) and (7) can be used in conjunction with Eqs. (3) and (4) to determine which strategy is more effective in increasing the signal-to-noise ratio.

As an example, the present system, based on the 68HC11 microcontroller, has a maximum bandwidth of 100 kHz and samples 100 voltage levels in a nominal sweep time of 0.1 s. A programmable software delay loop, which provides sufficient time for the system to settle following a change in the probe voltage, sets the actual bandwidth. Typical time scales are

$$t_o : 1 \text{ ms}, \quad t_w : 5\ \mu\text{s}, \quad t_r : 7.5\ \mu\text{s}, \quad t_c : 10\ \mu\text{s}, \quad t_d : 10\ \mu\text{s}.$$

Based on these time scales, the number of samples at each level, $n = 28$ while the time available for integration, $T = 0.97$ ms. Using these numbers in Eqs. (3) and (4) shows that the integrator increases the signal-to-noise ratio by 10.3 dB compared with the discrete averaging system. If the averaging system were to have the same SNR as the integrator then Eq. (5) indicates that some 300 samples would be required compared with the 28 attainable. The integrate-and-dump is therefore superior in this application in its ability to suppress noise. The SNRs for the two systems become comparable only if the I&D is operating near its minimum integration time of 0.1 ms.

If two ADCs are employed in the sampled system then n rises to 39, while if the sampled system is not limited by the ADCs then $n = 66$. Hence, even under these conditions, the I&D, with effectively 300 samples, remains competitive.

If very large voltage steps are employed, the delay time needs to be increased in order to provide sufficient time (30–60 μs) for the probe voltage to settle. This leads to a consequent decrease in the system's bandwidth. In this situation, some increase in the total sweep time is necessary if the SNR is to be maintained.

V. STAND-ALONE CONTROL

The probe driver operates on a standard-alone basis under the control of the microprocessor-based interface board. This board was developed in our laboratory and is employed as a standard interface for a variety of instruments and control functions.[10] The board, shown in block form in Fig. 7, features eight multiplexed analog inputs, two analog outputs, and 12 digital input/output lines. The board connects to the control PC via the standard industrial GPIB interface bus. Simple commands are transmitted over the GPIB to request measurements on the probe: either single point measurements at a set probe voltage or sweep measurements of a specified number of points over a specified voltage range. The current and voltage measurements are stored in on-board RAM and then down loaded to the control PC on request. The use of the interface board serves to decentralize the various control and measurement systems attached to the plasma

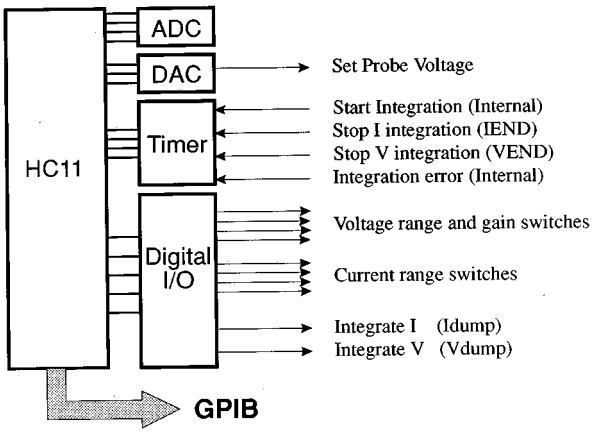


FIG. 7. Block diagram of the 68HC11-based interface board.

rig. This leads to far greater flexibility and eliminates the need for a separate dedicated PC for each instrument.

Operation: The driver has two modes of operation: single point and sweep, either of which is initiated by the following GPIB commands.

(1) ***P±VVV*** Take a measurement with a probe potential of $\pm VVV$ volts. The values of the actual probe potential and probe current are returned in the format $\pm vvv.v \pm iii.iii$ in volts and milliamps, respectively.

(2) ***S±VVV±FFF nn*** Sweep the probe potential from $\pm VVV$ to $\pm FFF$ volts in nn steps taking a measurement of probe potential and current at each point and storing the values in RAM.

(3) ***D nn*** Down load the first nn measurements from RAM in the format $nn\{\pm vvv.v \pm iii.iii\}$.

The probe is always parked at -125 V to avoid a build up of deposits on the probe tip whenever it is not making a measurement. If a single point is requested, the required voltage is set via the DAC of the microcontroller. The resistors of the current measurement block are switched to the most sensitive range and a measurement taken. They are then sequentially switched to less sensitive ranges until the overload flag clears. At this point, the $Idump$ and $Vdump$ switches are opened and the integration starts. The microcontroller times the integration which stops when the output of the integrator reaches a preset level as signalled by the two lines $IEND$ and $VEND$. The timer has a preset time out. If the level has not been reached in this time then the voltage (or current) is out of range and will be set to zero. The current and voltage are calculated from the timers, and the values returned to the control computer via the GPIB.

A similar sequence is initiated in sweep mode except that a series of values are taken and stored in RAM. The integrator time constants are set such that the maximum integration time for a point will be $\sim$1 ms. Thus a full sweep (including time overheads for range adjustments) can be completed in $<$150 ms.

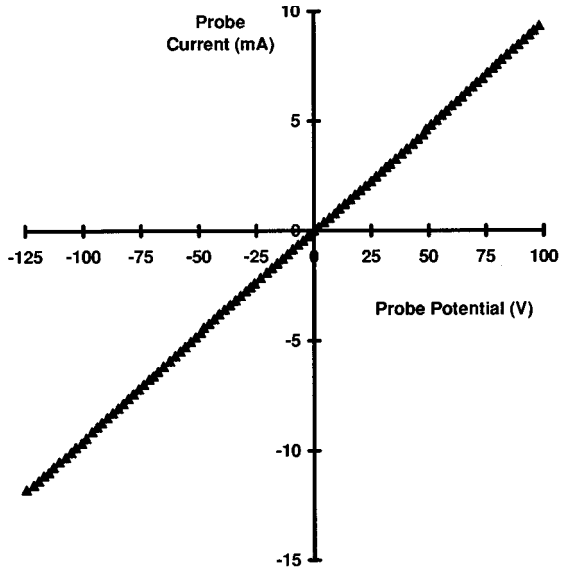


FIG. 8. Current–voltage characteristic obtained by the system using a 11 kΩ resistor as the load.

VI. TESTING AND EVALUATION

In the initial test, the plasma was replaced by a 11 kΩ resistor and a 100 step sweep covering the range from -100 to $+100$ V was acquired. Figure 8 shows the trace of current as a function of voltage that was obtained. This shows the expected linear response with a slope of 10.58 kΩ compared with a meter value of 10.93 kΩ. Measurements with other resistors gave a systematic discrepancy of 2.8% $+/-$0.3%. Fine tuning of the sense resistors should reduce this error. The maximum nonlinearity was found to be $<$1% while the zero offset error was <3 μA for currents in the range from 0 to 10 mA.

The probe driver was then connected to a Langmuir probe inserted into a helicon wave plasma source heated by a

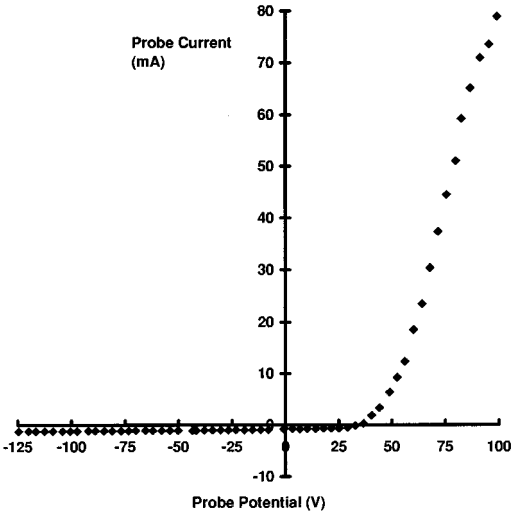


FIG. 9. Current–voltage characteristic for a Langmuir probe inserted in a helicon wave plasma source.

1.25 kW rf generator operating at 13.56 MHz. Figure 9 shows the probe's I–V characteristic obtained in one sweep of 100 points between −125 and +100 V in 0.1 ms. Little detectable noise has been recorded by the system despite the presence of a large amount of rf radiation in the vicinity of the probe. However, the actual shape of the characteristic is not correct due to the effects of the rf on the behavior of the probe. An exponential fit to the data in the vicinity of the knee of the characteristic yields an apparent electron temperature of 6 eV compared with the 3 to 4 eV typical of helicon sources.[11] In order to eliminate this problem, some form of high impedance choke built into the probe is required as described by several authors.[6,12,13]

VII. DISCUSSION

This article has described the design, implementation, and testing of a microcontroller-based Langmuir probe driver. Novel features of the system include:

(1) the use of integrate-and-dump techniques to implement the process of analog-to-digital conversion and to provide effective noise suppression,
(2) an efficient strategy for automatic range changing of both the probe current and voltage measurements,
(3) a solution to the floating power supply problem based on the PSRR characteristics of a high voltage operational amplifier, and
(4) a microcontroller-based interface and control board employing the GPIB communication protocol.

Using these features, the system has achieved its design specifications, and has been successfully demonstrated in the noisy environment of a helicon plasma source. In particular, it has been demonstrated that the system can provide smoothed current and voltage measurements at 100 data points as it sweeps any voltage range between −125 and +100 V in a time <0.15 s.

Future work centers on the suppression of rf effects on the probe characteristic possibly using high impedance chokes built into the probe and automatic readout of the plasma density and electron temperature.

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